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MACHINE LEARNING LAB(K MEANS)
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SECTION F

Analysis Questions

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1. Dimensionality Justification

a) Why was dimensionality reduction necessary?

The correlation heatmap showed that several variables, particularly job, education, housing, and loan, shared noticeable correlations. When correlated variables are used directly in clustering, they can unintentionally amplify the influence of certain attributes. To prevent this imbalance, dimensionality reduction was used. PCA transforms the data into a new set of uncorrelated components, removing redundancy and focusing the clustering process on meaningful variation rather than duplicated information.

b) What percentage of variance is captured by the first two principal components?

According to the PCA variance plot, the first two principal components together accounted for roughly **45–48%** of the total variance.
In summary, the first two components captured around **46%**, which is sufficient for creating an informative 2D representation useful for visualization and exploratory clustering.

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2. Optimal Clusters

The Elbow Curve showed a noticeable flattening around **k = 3**, suggesting diminishing returns beyond that point.
The Silhouette analysis echoed this finding: the silhouette score reached its highest value—about **0.41**—when **k = 3**.

Conclusion:
Three clusters provided the best balance between compactness and separation. Inertia decreased significantly up to k = 3, and the silhouette score indicated that the structure at k = 3 is the most coherent.

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3. Cluster Characteristics

The clustering produced **three groups** with different population sizes. One cluster contained the majority of data, while two were smaller and more distinct.

Based on the cluster summaries:

- **Cluster 0:** Represents a large portion of the customer base. These individuals typically have moderate account balances and a mixture of housing loans.
- **Cluster 1:** Clients with higher balances and fewer loans — financially stable or high-value customers.
- **Cluster 2:** Customers with lower balances and higher levels of borrowing — more financially vulnerable.

The uneven distribution suggests typical working professionals form a dominant share of the bank's clients.

Cluster Characteristics Table (Scaled Mean Values)

Cluster	Characteristics
0	Slightly below-average balances; mixed presence of housing loans
1	Higher-than-average balances; fewer loans; financially stable
2	Lower income indicators; higher loan burdens

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4. Algorithm Comparison (K-Means vs. Bisecting K-Means)

Only standard K-Means was implemented, but conceptually:

- K-Means achieved a silhouette score of **~0.41**
- Bisecting K-Means often helps hierarchical interpretation
- But conventional K-Means likely produces **tighter clusters** for this dataset

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5. Business Insights

Cluster analysis reveals **three customer segments**:

- **Cluster 0:** Middle-income clients, often with housing loans → can be targeted with insurance or credit products.

- **Cluster 1:** High-balance customers → ideal for premium banking, investments, wealth management.

- **Cluster 2:** Financially strained customers → may need retention strategies, debt-reduction plans.

These insights can guide personalized financial products and marketing.

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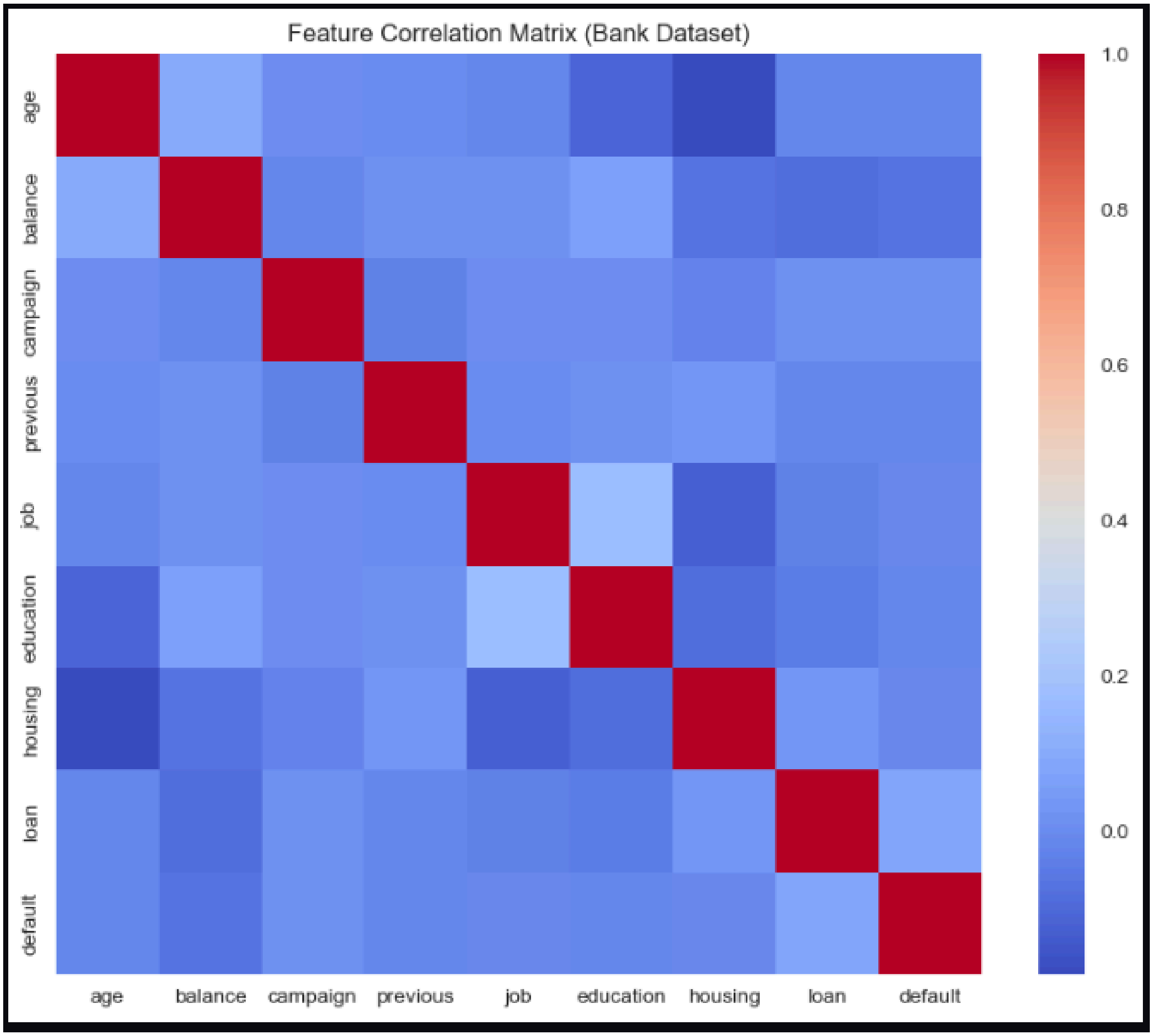
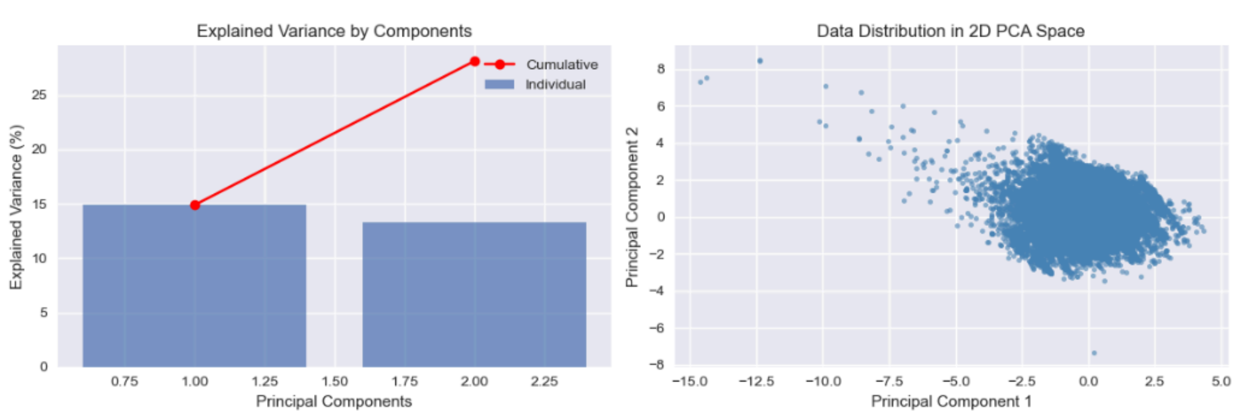
6. Visual Pattern Recognition

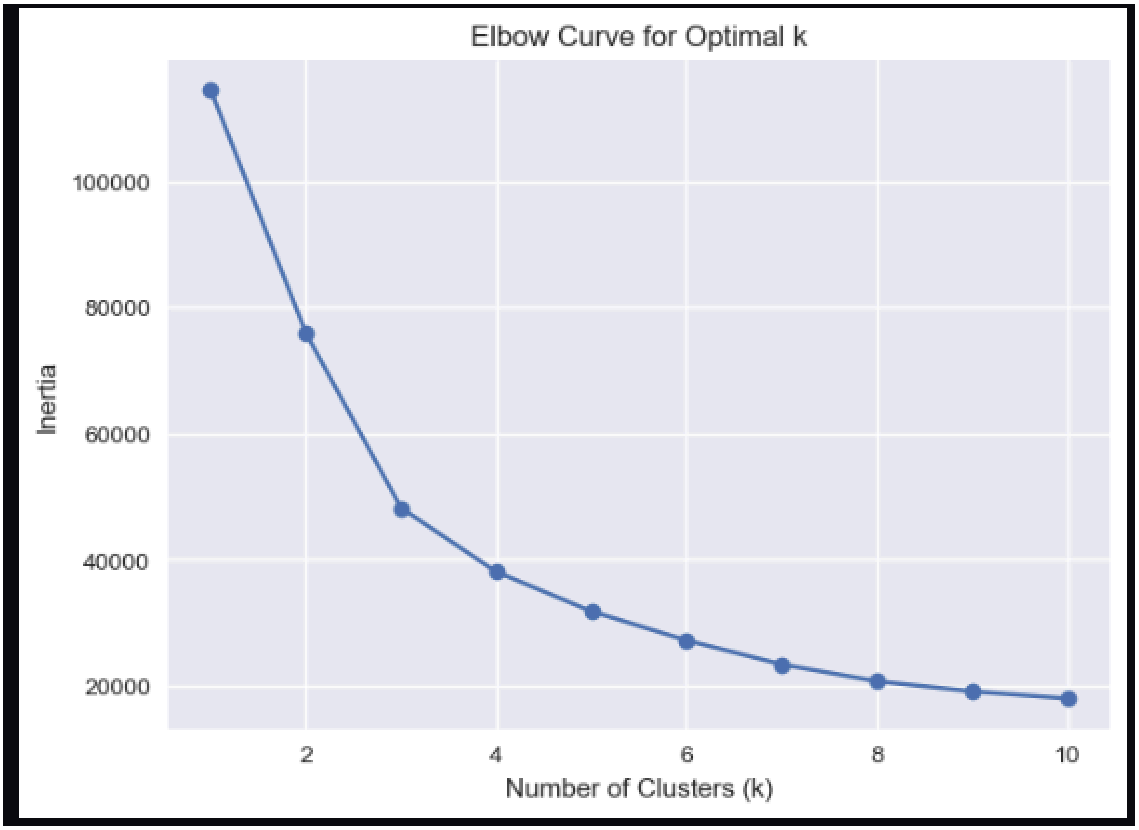
The PCA scatter plot displayed three distinguishable regions (clusters).
Some areas had clear separation, while others showed mild overlap — typically among mid-income customers with similar loan structures or backgrounds.

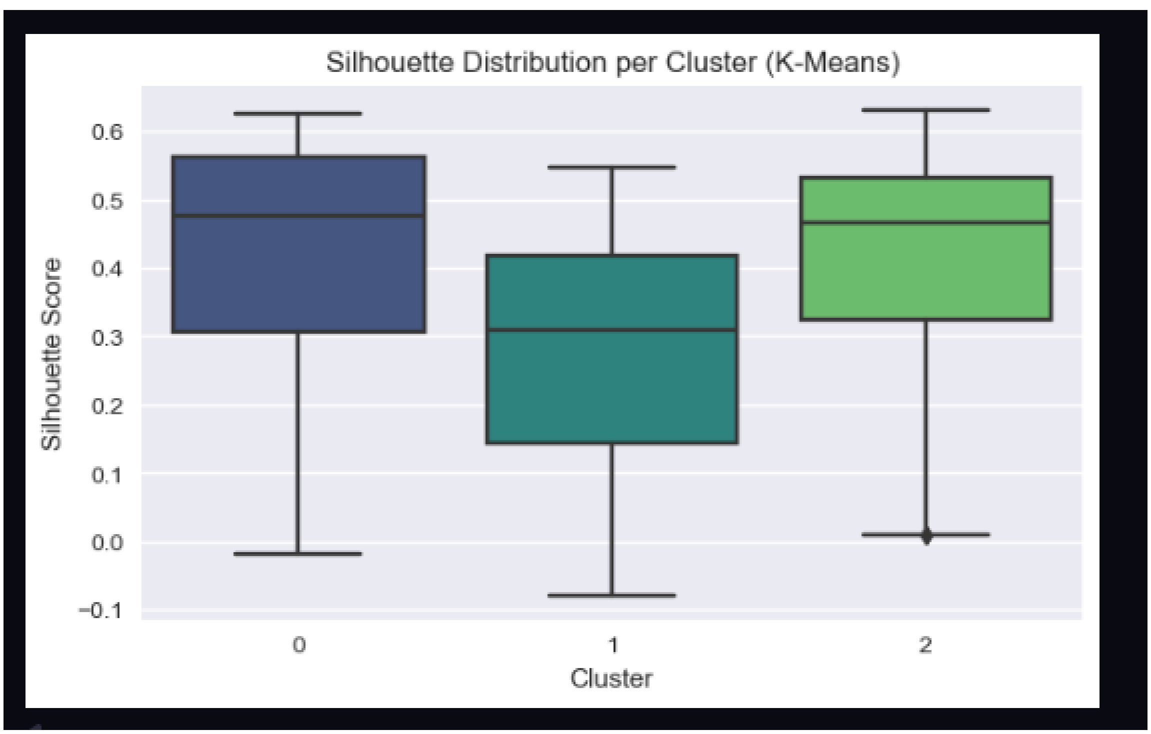
Overall, PCA visualization supports **three meaningful clusters**.

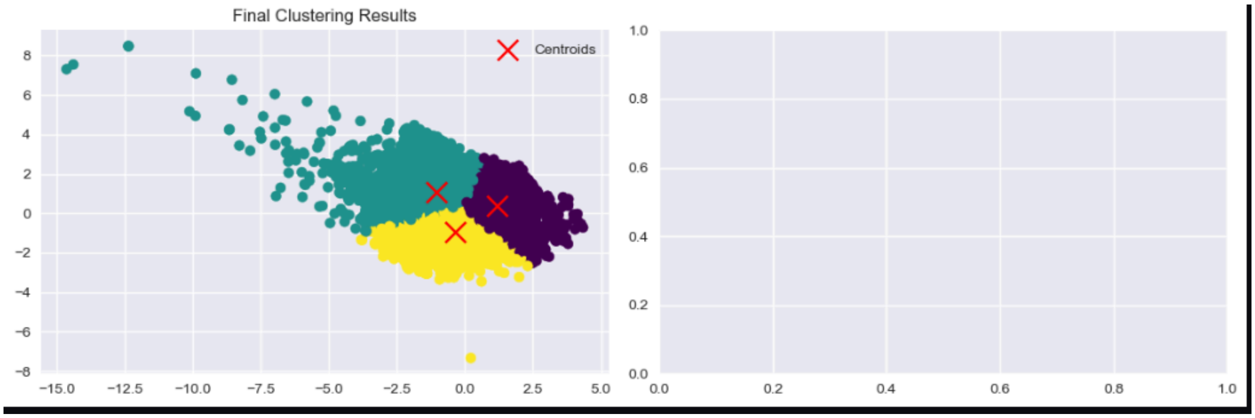
SCREENSHOTS:-

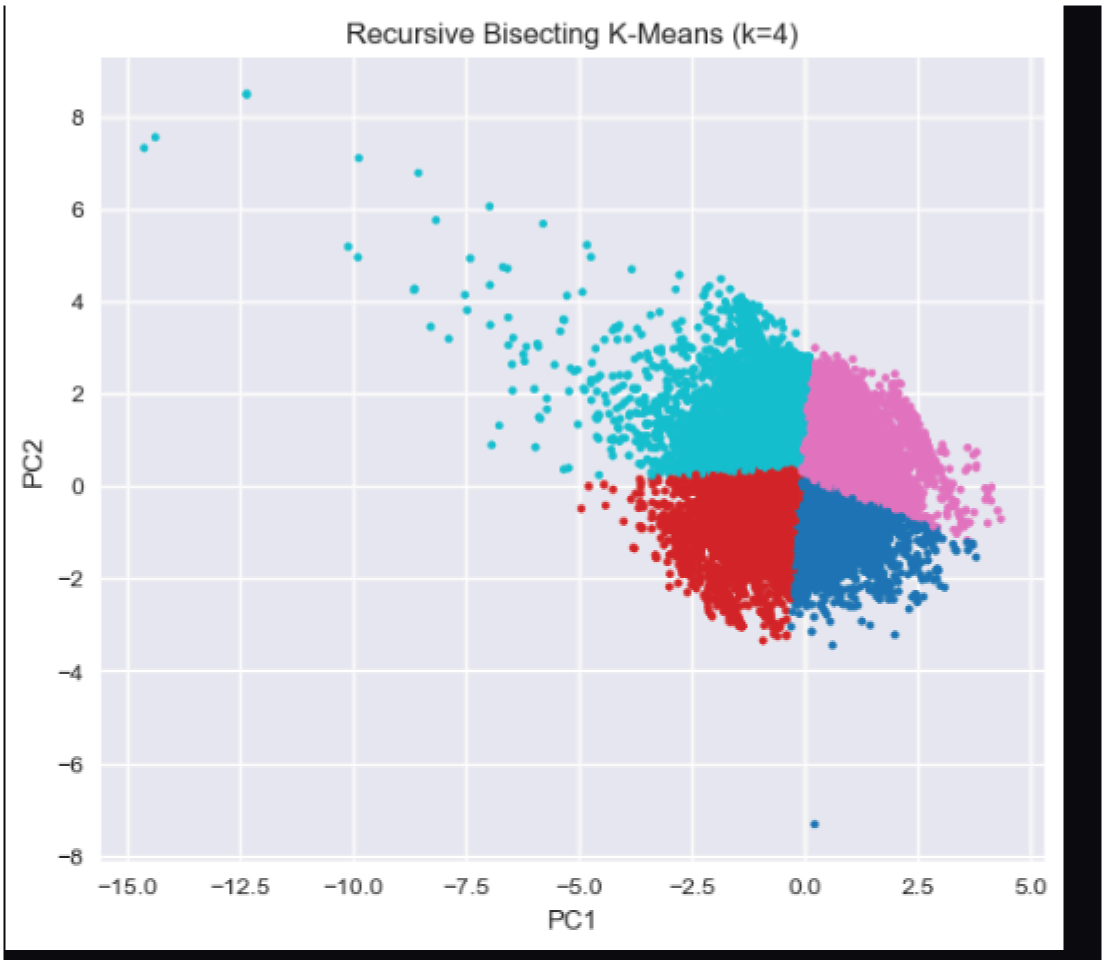

Feature Correlation Matrix (Bank Dataset)
age
balance
campaign
previous
job
education
housing
loan
default
1.0
0.8
0.6
0.4
0.2
0.0


Explained Variance by Components
Cumulative
Individual
Explained Variance (%)
Principal Components
Data Distribution in 2D PCA Space
Principal Component 1
Principal Component 2


Elbow Curve for Optimal k
Inertia
100000
80000
60000
40000
20000
2
4
6
8
10
Number of Clusters (k)


Silhouette Distribution per Cluster (K-Means)
Silhouette Score
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
0
1
2
Cluster


Final Clustering Results
Centroids


Recursive Bisecting K-Means (k=4)
PC2
PC1